
Tobechukwu Charles-Ibe

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Education

University Of St Andrews – BSc Computer Science

2022 - Expected 2026

Expected Grade: First Class

High School Of Dundee

2017 – 2022

Advanced Higher: Mathematics (A), Physics (A), Computing Science (B)

Higher: Physics(A), Computing Science(A), Mathematics(A*), Biology(A), Chemistry(A)

Skills

- **Programming Languages:** Python, Haskell, C, Java, JavaScript
- **Tools & Technologies:** Git, HTML, React, Tailwind CSS, SQL, Various Other JS Libraries

Notable Projects

PORTFOLIO WEBSITE

April, 2024

- Developed a personal [portfolio website](#) that showcased skills and experience with web development using React.js and Tailwind CSS
- Implemented a 'Contact Me' page integrated with the emailJS API, streamlining communication between the users of the site by facilitating automated email responses directly from the website.
- Integrated other JS libraries for advanced additional functionality such as smooth scrolling, page routing, text animations, notifications and interactive backgrounds, enriching the end user experience.

C++ MEMORY ALLOCATOR

November, 2024

- Constructed a Buddy System Memory Allocator for the StACSOS operating system
- Implemented a System to allow processes to request a number of memory pages in memory from the Operating System and allocated pages whilst keeping track of free pages
- Optimised memory allocation in the system and reduced external fragmentation in memory

JAVA CHATBOT

March, 2023

- Developed a Java chatbot inspired by the Eliza chatbot, using Regex and Pattern Matching to process user interactions
- Implemented functionality to load in different conversation scripts to allow chatbot to "play different roles"
- Designed and implemented a system of pre and post substitutions to understand and respond better to user inputs
- Incorporated random response selection features and realistic delays to make interactions more natural and engaging

Leadership & Certifications

GOLDMAN SACHS SOFTWARE ENGINEERING VIRTUAL EXPERIENCE | FORAGE

- Completed a job simulation as a Goldman Sachs governance analyst responsible for assessing IT security and suggesting improvements.
- Identified that the company was using an outdated password hashing algorithm by cracking passwords using Hashcat.
- Wrote a memo for my supervisor summarizing a range of proposed uplifts to increase the company's level of password protection including extending minimum password length and using a dedicated hashing algorithm.

RUN CLUB VICE PRESIDENT AND CO FOUNDER | RUNWAY ST ANDREWS

- Co-founded a run club in the University town of St Andrews
- Organised new members of the run club together into different pace groups and organised weekly runs and their routes and helped grow the run club to over 30 members

NATIONAL 5 MATHS MENTORING | HIGH SCHOOL OF DUNDEE

- Tutored a class National 5 Mathematics class, helping students with their problem-solving skills following the impact of the pandemic
- Provided goal oriented support and helped increase student's predicted National 5 grades

Awards

- United Kingdom Mathematics Trust (UKMT) Silver Award
- Duke Of Edinburgh Silver Award

Additional Information

Interests: Programming | Artificial Intelligence | Web Development | Cooking | Networking | Education | New Technologies | Poker Fan | Member of University of St Andrews Football Team | Arsenal Fan | Afro-Caribbean Society Member | Run Club Founder